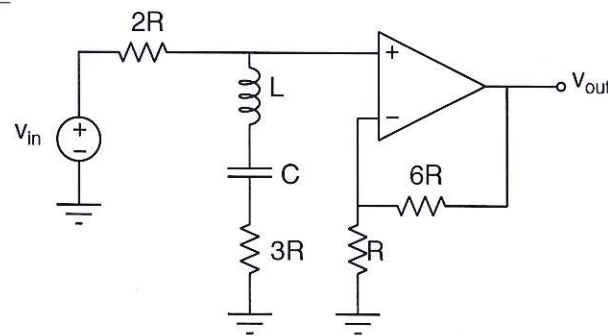


Question 1 [25 Points]:



$R = 100 \Omega$
announced in class

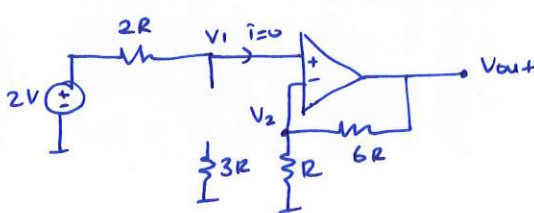
For the circuit above, $L = 2 \text{ mH}$ and $C = 5 \text{ nF}$. Assume that the OPAMP is not saturated.

(a) [7 points] Find v_{out} if $v_{in} = 2 \text{ V}$ (DC).

For DC, capacitor $\rightarrow$ open, inductor $\rightarrow$ short

$$V_1 = V_2 = 2 \text{ V}$$

$$V_2 = V_{out} \cdot \frac{R}{R + 6R} \Rightarrow V_{out} = 7 V_2 = \boxed{14 \text{ V}}$$



(b) [5 points] What is the resonance frequency, f_0 , of the LC circuit in Hz?

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \boxed{50.3 \text{ kHz}}$$

(c) [5 points] What is the Q of the LC tuned circuit? What is its 3-dB bandwidth in Hz?

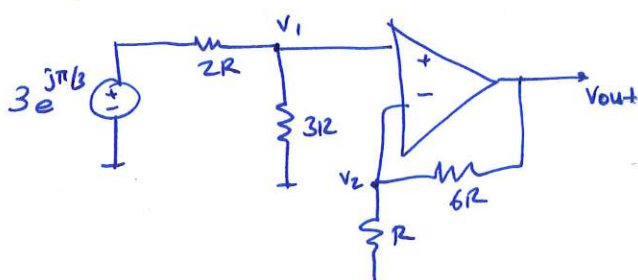
Series tuned circuit.

$$Q = \frac{\omega_0 L}{R_s} = \frac{2\pi f_0 L}{2R + 3R} = \frac{2\pi \times 50.3 \times 10^3 \times 2 \times 10^{-3}}{5 \times 10^2} = \boxed{1.26}$$

$$\Delta f = \frac{f_0}{Q} = \frac{50.3 \text{ kHz}}{1.26} = \boxed{39.9 \text{ kHz}}$$

(d) [8 points] Find $v_{out}(t)$ if $v_{in}(t) = 3 \cos(\omega_0 t + \pi/3)$ Volts, with ω_0 at resonance frequency.

At resonance, series tuned circuit acts like short circuit.



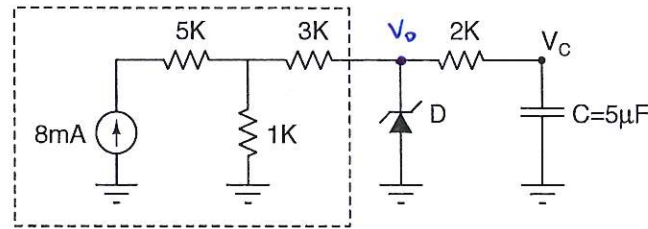
$$V_1 = V_{in} \cdot \frac{3R}{2R + 3R} = \frac{3}{5} V_{in}$$

$$V_2 = V_{out} \cdot \frac{R}{R + 6R} = \frac{V_{out}}{7}$$

$$V_1 = V_2 \Rightarrow V_{out} = \frac{21}{5} V_{in}$$

$$V_{out}(t) = \boxed{12.6 \cdot \cos(\omega_0 t + \pi/3) \text{ Volts}}$$

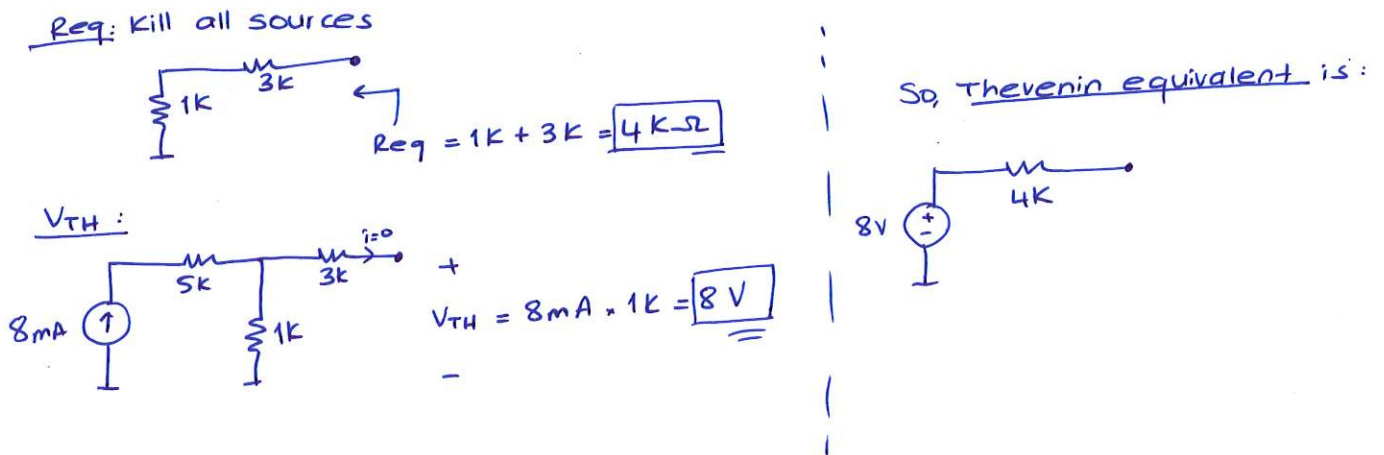
Question 2 [25 Points]:



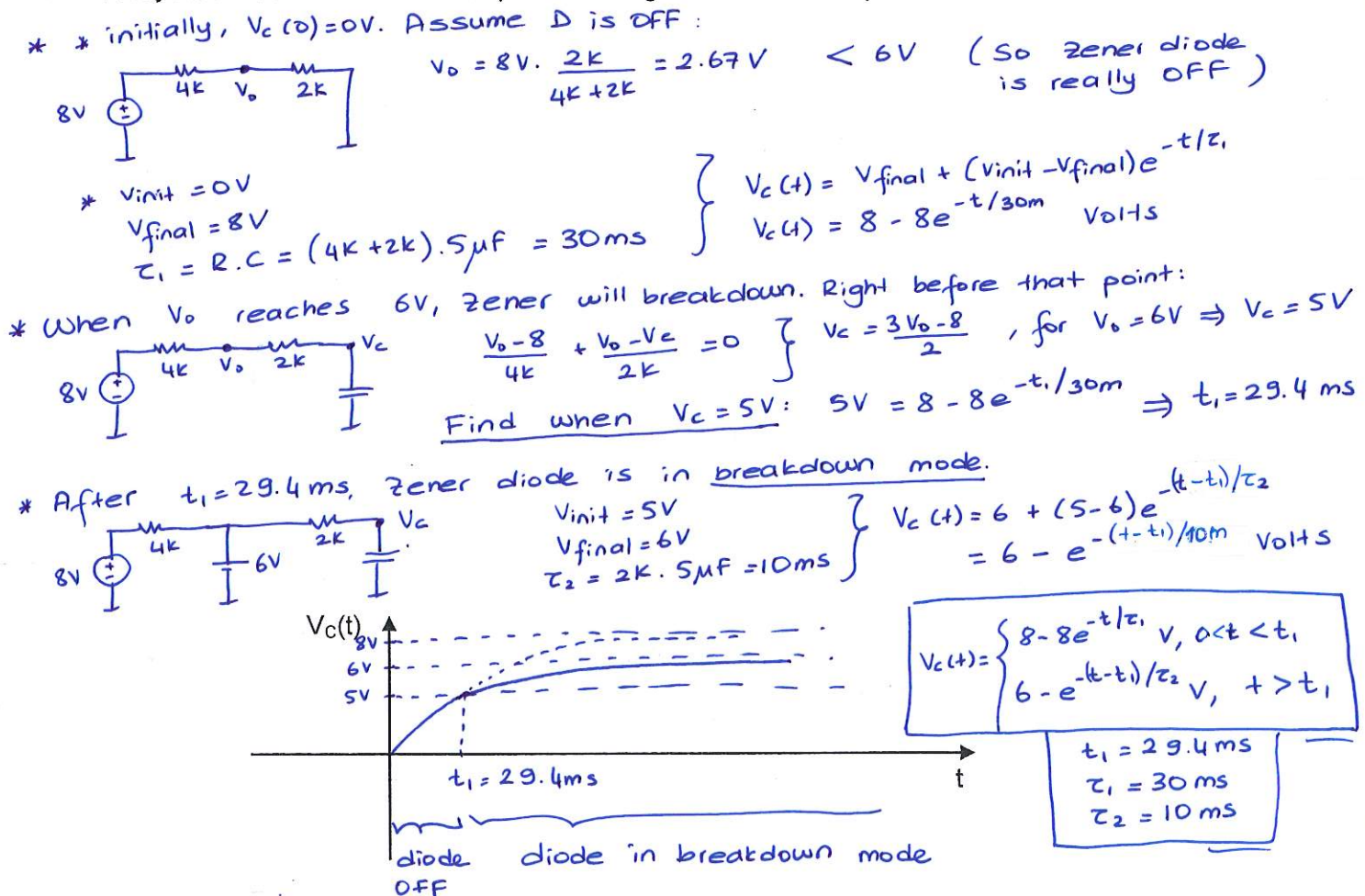
In the circuit above, the breakdown voltage for Zener diode D is $V_Z = 6V$.

For the capacitor, $V_C(0) = 0V$.

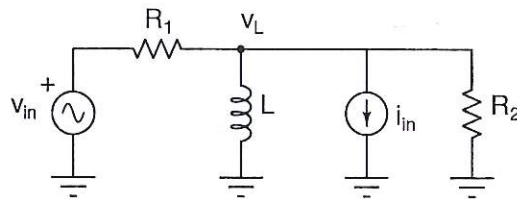
(a) [7 points] Find and draw the Thevenin equivalent of the circuit inside the dashed box.



(b) [18 points] Find and plot $V_C(t)$ for $t > 0$. Clearly specify the state of the diode for all $t > 0$. Plot your result below. Mark all important voltage levels and time points.



Question 3 [25 points]:



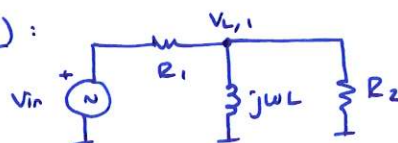
In the circuit above, $R_1 = 1\text{K}\Omega$, $R_2 = 3\text{K}\Omega$, $L = 0.5\text{ mH}$.

Find the voltage across the inductor, $v_L(t)$, if the voltage and current sources are

$$v_{in}(t) = 4\sqrt{2} \cos\left(15 \cdot 10^5 t + \frac{\pi}{4}\right) \text{ Volts, and } i_{in}(t) = 8 \cos(\sqrt{3} \cdot 15 \cdot 10^5 t) \text{ mA.}$$

Inputs at different frequencies $\Rightarrow$ use superposition!

* From V_{in} (kill I_{in}):



$$V_{in} = 4\sqrt{2} e^{j\pi/4}$$

$$j\omega L = j15 \cdot 10^5 \cdot 0.5 \times 10^{-3} = j750\Omega = j0.75\text{ k}\Omega$$

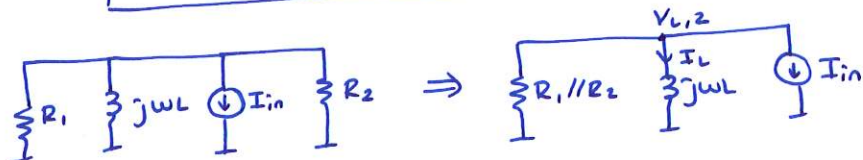
From voltage divider:

$$V_{L,1} = V_{in} \cdot \frac{(j\omega L \parallel R_2)}{R_1 + (j\omega L \parallel R_2)} = V_{in} \cdot \frac{\frac{j\omega L \cdot R_2}{j\omega L + R_2}}{R_1 + \frac{j\omega L \cdot R_2}{j\omega L + R_2}} = V_{in} \cdot \frac{j\omega L R_2}{R_1 R_2 + j\omega L (R_1 + R_2)}$$

$$= 4\sqrt{2} e^{j\pi/4} \cdot \frac{j0.75\text{ k} \cdot 3\text{ k}}{1\text{ k} \cdot 3\text{ k} + j0.75\text{ k} \cdot 4\text{ k}} = 4\sqrt{2} e^{j\pi/4} \cdot \frac{3j}{4 + 4j} = 4\sqrt{2} e^{j\pi/4} \cdot \frac{3e^{j\pi/2}}{4\sqrt{2} e^{j\pi/4}}$$

$$V_{L,1} = 3e^{j\pi/2} \Rightarrow \boxed{v_{L,1}(t) = 3 \cos(15 \cdot 10^5 t + \pi/2) \text{ Volts}}$$

* From I_{in} (kill V_{in}):



$$I_{in} = 8 \times 10^{-3}$$

$$j\omega L = j\sqrt{3} \cdot 15 \cdot 10^5 \cdot 0.5 \times 10^{-3} = j\sqrt{3} \cdot 750\Omega = j\sqrt{3} \cdot 0.75\text{ k}\Omega$$

$$I_L = -I_{in} \cdot \frac{R_1 \parallel R_2}{R_1 \parallel R_2 + j\omega L} = -8 \times 10^{-3} \cdot \frac{0.75\text{ k}}{0.75\text{ k} + j\sqrt{3} \cdot 0.75\text{ k}} = -\frac{8 \times 10^{-3}}{1 + j\sqrt{3}}$$

^ (current divider)

$$V_{L,2} = I_L \cdot j\omega L = -\frac{8 \times 10^{-3}}{1 + j\sqrt{3}} \cdot j\sqrt{3} \cdot 0.75\text{ k} = \frac{e^{-j\pi/2} \cdot 6\sqrt{3}}{2 e^{j\pi/6}} = 3\sqrt{3} e^{-j5\pi/6}$$

$$\boxed{v_{L,2}(t) = 3\sqrt{3} \cos(\sqrt{3} \cdot 15 \cdot 10^5 t - 5\pi/6) \text{ Volts}}$$

* In total:

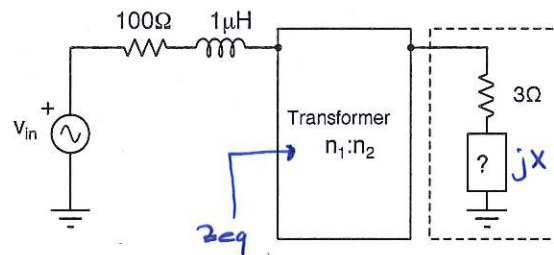
$$v_L(t) = v_{L,1}(t) + v_{L,2}(t)$$

$$\boxed{v_L(t) = 3 \cos(15 \cdot 10^5 t + \pi/2) + 3\sqrt{3} \cos(\sqrt{3} \cdot 15 \cdot 10^5 t - 5\pi/6) \text{ Volts}}$$

OR

$$v_L(t) = -3 \sin(15 \cdot 10^5 t) + 3\sqrt{3} \cos(\sqrt{3} \cdot 15 \cdot 10^5 t + 7\pi/6) \text{ Volts}$$

Question 4 [25 Points]:



We have a source with internal impedance represented by a 100Ω resistor in series with $1\mu\text{H}$ inductor. $V_{in}(t) = 5 \cos(2\pi ft)$ Volts, where $f = 1 \text{ Mhz}$. The dashed box represents our load, which already contains a 3Ω resistor. To maximize the power delivered to the load, we need to choose an appropriate reactive component in place of "?" and also design a transformer in between.

Important: For transformer, we will use a core with $A_L = 6 \text{ nH/turns}^2$. Use the closest integer values for the number of turns.

- (a) [7 points] What reactive component should we put in place of "?", and what should be its value?

$$Z_L = 3 + jX \quad / \quad Z_{eq} = \left(\frac{n_1}{n_2}\right)^2 Z_L = \left(\frac{n_1}{n_2}\right)^2 (3 + jX)$$

For max. power transfer: $Z_{eq} = Z_s^* = (100 + j2\pi \cdot 10^6 \cdot 10^{-6})^* = 100 - j2\pi$

$$\left(\frac{n_1}{n_2}\right)^2 (3 + jX) = 100 - j2\pi \quad \left\{ \begin{array}{l} \text{Real and imaginary parts must match on} \\ \text{both sides.} \end{array} \right.$$

$$\left(\frac{n_1}{n_2}\right)^2 \cdot 3 = 100 \Rightarrow \boxed{\frac{n_1}{n_2} = \sqrt{\frac{100}{3}}} \quad \text{Then, } \left(\frac{n_1}{n_2}\right)^2 X = -2\pi \quad (\text{this is a capacitor!!})$$

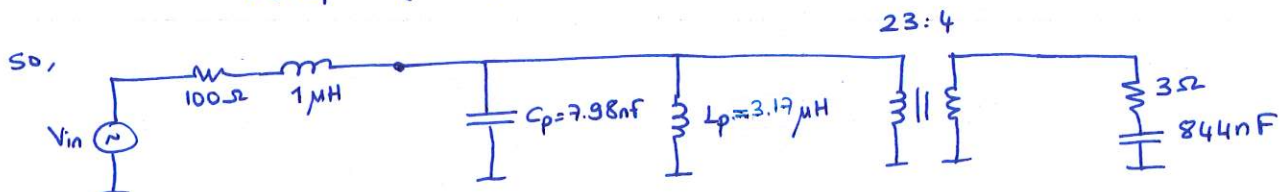
$$\frac{100}{3} \cdot \frac{1}{j\omega C} = -j2\pi \Rightarrow C = \frac{100}{2\pi \cdot 2\pi \cdot 10^6 \cdot 3} = \boxed{844 \text{ nF}}$$

- (b) [13 points] Design a narrowband transformer. Specify the number of turns and values for all the necessary components. Draw the entire circuit.

$$\frac{n_1}{n_2} = \sqrt{\frac{100}{3}} \approx 5.77 \quad / \quad \boxed{\begin{matrix} n_1 = 23 \\ n_2 = 4 \end{matrix}} \quad \text{gives } \frac{n_1}{n_2} = 5.75 \quad (\text{very close!})$$

$$L_p = n_1^2 A_L = 23^2 \cdot 6 \text{ nH} = \boxed{3.17 \mu\text{H}}$$

$$C_p = \frac{1}{\omega_0^2 L_p} = \frac{1}{(2\pi \cdot 10^6)^2 \cdot 3.17 \times 10^{-6}} = \boxed{7.98 \text{ nF}}$$

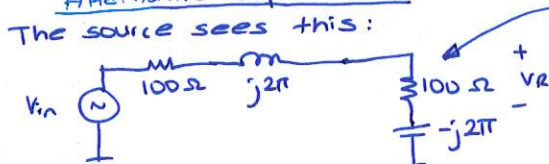


- (c) [5 points] What is the power delivered to the load for the maximum power transfer case?

We get all the power that the source can deliver
(i.e., power available)

$$P_d = \frac{V_s^2}{8R_s} = \frac{5^2}{8 \cdot 100} = \boxed{31.2 \text{ mW}}$$

Alternative explanation:



load resistor dissipates the power

$$V_R = \frac{V_{in}}{2}$$

$$P_d = \frac{1}{2} \frac{V_R^2}{R} = \frac{1}{2} \left(\frac{V_{in}}{2}\right)^2 \cdot \frac{1}{R} = \frac{V_{in}^2}{8R} = \frac{5^2}{8 \cdot 100}$$

$$= \boxed{31.2 \text{ mW}}$$